



Unifying Graph Out-of-Distribution Generalization and Detection through Spectral Contrastive Invariant learning

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Abstract

Graph representation learning encounters great difficulties under distribution shifts. This challenge has aroused considerable interest in graph out-of-distribution (OOD) generalization and detection, which can effectively handle covariate and semantic shifts, respectively. However, real-world graph tasks often involve complex unlabeled wild data with both covariate and semantic shifts, motivating a critical question: *can we design a unified framework for joint graph OOD generalization and detection?* Invariant graph learning, which extracts stable relationships between features and labels, offers a promising candidate for joint OOD generalization and detection, but faces three critical challenges (1) how to model invariant subgraphs with unlabeled data, (2) how to ensure graph representations benefit both tasks, and (3) how to integrate labeled and unlabeled data under proper invariance principles. To solve these challenges, we introduce Unified Graph Out-Of-Distribution generalization and detection framework (**UniGOOD**) with three tailored components. Specifically, to capture subgraphs without relying on labels, we first propose the distributional invariant subgraph generator to model subgraph conditional distributions. Next, to enable generalization and reliable detection, we propose the cross-invariant-subgraph spectral contrastive learning module to learn invariant representations from subgraph distributions. Finally, for accurate subgraph discovery across labeled and unlabeled graphs, we design the triple-population invariance regularizer to enforce the invariance principle through spectral graph theory. We prove that our method theoretically ensures accurate invariant subgraphs, enabling effective OOD generalization and detection. Experiments

show that **UniGOOD** outperforms state-of-the-art baselines for both graph OOD generalization and detection tasks.

CCS Concepts

• **Mathematics of computing** → **Graph algorithms**; • **Computing methodologies** → **Neural networks**.

Keywords

Out-of-Distribution Generalization, Out-of-Distribution Detection, Graph Machine Learning

ACM Reference Format:

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Resource Availability:

The source code of this paper has been made publicly available at <https://doi.org/10.5281/zenodo.18327040> and can also be accessed via the project repository: <https://github.com/Lowy999/UniGOOD>.

1 Introduction

Graph representation learning has become a critical and rapidly evolving field [45], achieving impressive results in various domains, such as modeling molecules [44], social networks [9], and knowledge graphs [3]. For example, graph neural networks (GNNs) [41] design end-to-end learning schemes to extract useful graph information through message-passing. However, existing graph representations learning approaches often confront graph data distribution shifts, i.e., the training and testing graph from different distributions. Graph data distribution shifts widely exist [16], stemming from the inherent complexity of graph generation processes and fundamentally impacts high-stakes graph applications including but not limited to medical diagnosis [18], financial risk assessment [42], and molecular property prediction [17, 40].

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To address the distribution shift challenge, recent research has developed two categories of approaches: graph OOD generalization [16] and graph OOD detection [4]. Graph OOD generalization methods primarily focus on handling covariate shifts [4], i.e., the distribution of input has been shifted between training and test data [11]. The key objective of generalization is to achieve satisfactory performance on test data under unknown distribution shifts. On the other hand, graph OOD detection methods specialize in identifying semantic shifts, where the fundamental meaning or class associations of features may change. These methods aim to reliably distinguish in-distribution (ID) samples from OOD samples, enabling the detection of OOD inputs.

Despite advances, current graph OOD generalization and detection methods remain separate, i.e., specializing exclusively in either generalization under covariate shifts or detection of semantic shifts. But real-world graph tasks usually involve complex unlabeled wild data with both covariate and semantic shifts. The inability of existing approaches to handle such multifaceted distribution shifts has become a major bottleneck for graph representation learning. This fundamental limitation raises a critical research question:

Can we design a unified framework for joint graph OOD generalization and detection?

Invariant graph learning, which extracts invariant subgraphs that maintain stable relationships between features and labels across environments, offers a promising candidate for joint OOD generalization and detection with joint labeled and unlabeled wild data. In theory, optimal OOD generalization can be achieved when predictions rely exclusively on invariant features [16], while effective OOD detection similarly benefits from focusing solely on invariant patterns. Nevertheless, designing a joint OOD generalization and detection framework based on invariant graph learning faces three substantial technical challenges:

First, how to model invariant subgraphs with unlabeled data remains largely unexplored, as supervised methods [17, 39] require graph labels and cannot handle unlabeled wild data. Second, how to ensure that invariant graph representations can simultaneously enhance both OOD generalization and detection tasks remains unknown, especially under combined covariate and semantic shifts. Lastly, how to design invariance principles that integrate labeled and unlabeled wild data is still challenging, as supervised methods ignore unlabeled data while unsupervised ones lack label guidance.

To address these challenges, we propose **UniGOOD** (Unified Graph Out-Of-Distribution generalization and detection), a novel framework that captures invariant graph patterns under both covariate and semantic shifts while enabling OOD generalization and detection. As illustrated in Figure 1, our method can achieve this goal by jointly optimizing three complementary modules, each targeting one of the aforementioned challenges. To learn invariant subgraphs from joint labeled and unlabeled graphs, we first propose the distributional invariant subgraph generator, which employs a variational graph autoencoder (VGAE) based architecture to model invariant subgraph conditional distributions. Then, to ensure OOD generalization and detection, we propose the cross-invariant-subgraph spectral contrastive learning module to subsequently leverage these distributions to learn invariant representations. Finally, for accurate subgraph discovery for labeled and

unlabeled graphs, we design the triple-population invariance regularizer to enforce the invariance principle that identifies invariant subgraphs with stable label relationships through spectral graph theory. We theoretically prove that our proposed invariance principle ensures the method produces accurate invariant subgraphs, and that using these subgraphs for graph representation enables OOD generalization and detection. Extensive experiments on both synthetic and real-world graph benchmarks demonstrate that **UniGOOD** achieves significant performance improvements for both graph OOD generalization and detection tasks compared to state-of-the-art baselines.

Our contributions are summarized as follows:

- We propose **UniGOOD**, a novel framework that jointly addresses graph OOD generalization and detection under both covariate and semantic shifts. To the best of our knowledge, we are the first to unify these two critical graph OOD tasks.
- We design three tailored modules, including distributional invariant subgraph generator, cross-invariant-subgraph spectral contrastive learning, and triple-population invariance regularizer, that can distinguish invariant and variant subgraphs across joint labeled and unlabeled wild graphs.
- We theoretically show that **UniGOOD** achieves provable graph OOD generalization and detection guarantees.
- Extensive empirical results demonstrate the effectiveness of **UniGOOD** on various synthetic and benchmark datasets under both covariate and semantic shifts.

2 Notations and Problem Formulation

Notations. We consider the empirical training set $\mathcal{D}_l \cup \mathcal{D}_u$ as a union of labeled and unlabeled graphs. The labeled set $\mathcal{D}_l = \{G_i, Y_i\}_{i=1}^n$, where Y_i belongs to known class space $\mathcal{Y}$. We use $\mathcal{G}$ to denote the set of all input graphs. let $y : \mathcal{G} \rightarrow \mathcal{Y}$ denote the ground-truth labeling function. Following [2, 34], the unlabeled wild set $\mathcal{D}_u = \{G_i\}_{i=1}^m$ consists of ID, covariate OOD and semantic OOD data, where each graph G_i is drawn from a mixture distribution defined as follows.

Definition 1. *The marginal distribution of unlabeled wild data is:*

$$\mathbb{P}_u := (1 - \pi_c - \pi_s)\mathbb{P}_l + \pi_c\mathbb{P}_{co} + \pi_s\mathbb{P}_{se}, \quad (1)$$

where $\pi_c, \pi_s \in [0, 1]$ and $\pi_c + \pi_s \in [0, 1]$, $\mathbb{P}_l, \mathbb{P}_{co}$ and $\mathbb{P}_{se}$ represent the marginal distribution of ID, covariate-shifted OOD, and semantic-shifted OOD graphs, respectively.

We formulate the graph OOD generalization and detection under covariate and semantic shifts as:

Problem 1. *Given labeled set $\mathcal{D}_l$ and unlabeled set $\mathcal{D}_u$, the task is to learn (1) a graph predictor $g_\theta(\cdot) : \mathcal{G} \rightarrow \mathcal{Y}$ that minimizes*

$$\Pr_{G \sim \mathbb{P}_{co}} [y(G) \neq g_\theta(G)], \quad (2)$$

and (2) an OOD detector $h_\theta(\cdot) : \mathcal{G} \rightarrow \{\text{IN}, \text{OUT}\}$ that minimize

$$\Pr_{G \sim \mathbb{P}_l \cup \mathbb{P}_{co}} [h_\theta(G) \neq \text{IN}] + \Pr_{G \sim \mathbb{P}_{se}} [h_\theta(G) \neq \text{OUT}], \quad (3)$$

where IN denotes the graph is drawn from $\mathbb{P}_l$ or $\mathbb{P}_{co}$, and OUT denotes the graph is drawn from $\mathbb{P}_{se}$. Here, θ denotes the parameters of the shared model underlying both $g_\theta(\cdot)$ and $h_\theta(\cdot)$.

3 Method

We introduce **UniGOOD**, shown in Figure 1. We first introduce the distributional invariant subgraph generator, which models invariant subgraph probability distributions. Next, we present the cross-invariant-subgraph spectral contrastive learning module, which leverages these invariant subgraph distributions to achieve OOD generalization and detection. Finally, we describe the triple-population invariance regularizer, which ensures accurate invariant subgraph identification by enforcing invariant learning principles.

3.1 Distributional Invariant Subgraph Generator

The distributional invariant subgraph generator is designed to model subgraph distributions using joint labeled and unlabeled graphs. Unlike previous methods that rely on fixed vector representation of subgraphs, our invariant subgraph generator $\Phi(\cdot)$ learns the probabilistic distributions of invariant subgraphs for each input graph. This design is grounded in two important priors about unlabeled wild graph data: (1) the fundamental presence of noise in wild graph observations, and (2) deterministic single-subgraph approaches tend to induce overfitting in the invariant subgraph encoder for unlabeled graphs. By modeling distributions of potential invariant subgraphs, our approach is more flexible in handling wild graph data while mitigating overfitting risks.

Specifically, given an input graph instance G , we follow VGAE [15] and employ a dual GNN encoder to parameterize the distribution:

$$\mu = \text{GNN}_\mu(G), \log \sigma = \text{GNN}_\sigma(G), \quad (4)$$

where $\text{GNN}_\mu(\cdot)$ and $\text{GNN}_\sigma(\cdot)$ are two learnable GNNs. The node representations are sampled as: $\mathbf{Z} \sim \mathcal{N}(\mu, \sigma^2)$, where $\mathcal{N}(\cdot, \cdot)$ is the Gaussian distribution. Then, we use a mask matrix $\mathbf{M}$ to obtain the adjacency matrix of the invariant and variant subgraphs as follows:

$$\begin{aligned} \mathbf{M}_{i,j} &= \text{Sigmoid}(\mathbf{Z}_i^\top \cdot \mathbf{Z}_j), \\ \mathbf{A}_I &= \text{Top}_t(\mathbf{M} \odot \mathbf{A}), \mathbf{A}_V = \text{Top}_{1-t}((1 - \mathbf{M}) \odot \mathbf{A}), \end{aligned} \quad (5)$$

where $\text{Sigmoid}(\cdot)$ is the sigmoid function, $\mathbf{A}_I$ and $\mathbf{A}_V$ are the adjacency matrix of the invariant and variant subgraphs, respectively, $\text{Top}_t(\cdot)$ selects the top t -percentage of elements with the largest values, $\odot$ means the element-wise matrix multiplication.

A key distinction from conventional VGAE [15] lies in the unknown target invariant subgraphs, making the computation of reconstruction loss impossible and leaving Kullback-Leibler (KL) divergence $\mathcal{D}_{\text{KL}}(\cdot, \cdot)$ as the only viable loss component. We introduce β_{KL} to weight the KL divergence term. Intuitively, β_{KL} controls the strength of graph augmentation. A larger β_{KL} leads to greater diversity in the generated subgraphs. When $\beta_{\text{KL}} = +\infty$, the generator corresponds to randomly selecting edges for subgraph construction. Conversely, a smaller β_{KL} enhances the accuracy of the generated subgraphs. When $\beta_{\text{KL}} = 0$, the generator reduces to a standard Graph Auto-Encoder (GAE) [15] and produces identical subgraphs without augmentation. This augmentation based on KL divergence provides prior information for subgraph representations of unlabeled graphs while simultaneously mitigating model overfitting to training data. Besides, in Section 3.3, we will introduce the triple-population invariance regularizer to further optimize invariant subgraph identification.

3.2 Cross-Invariant-Subgraph Spectral Contrastive Learning

The module leverages invariant subgraph distributions of joint labeled and unlabeled graphs to achieve OOD generalization and detection through spectral contrastive learning [14]. Spectral contrastive learning utilizes spectral graph theory's normalized adjacency matrix to enable accurate linear classification even with limited unlabeled data, making it an effective approach for learning graph representations from unlabeled data.

To predict labels from invariant information of joint labeled and unlabeled data based on spectral contrastive learning, we first define the population graph. The population graph is a meta-level graph structure whose nodes correspond to all data instances in the input dataset, including both labeled and unlabeled examples. The edges of this graph encode semantically meaningful relationships between data instances. This hierarchical representation, analogous to the graph-of-graphs framework [35], allows us to capture global dataset-level patterns through spectral graph analysis.

Concretely, considering invariant subgraphs G_I for the label Y , we denote its population graph as $G_{\text{pop}}^{I \rightarrow Y} = (\mathcal{G}_I, \mathbf{w}^{I \rightarrow Y})$, where the node set $\mathcal{G}_I$ comprises all possible invariant subgraphs extracted from both labeled ID graphs $\mathcal{D}_I$ and unlabeled wild graphs $\mathcal{D}_u$, while $\mathbf{w}^{I \rightarrow Y}$ denotes the edge weights that will be specified subsequently. This population graph is designed such that invariant subgraphs from the same class exhibit stronger correlations, while those from different classes show weaker correlations.

We assume that if two invariant subgraphs are extracted from graphs of the same class, their corresponding nodes in $G_{\text{pop}}^{I \rightarrow Y}$ should have stronger edge connections, formally defined as follows [34]:

Definition 2. The overall similarity for any pair of data (G_I, G'_I) in $G_{\text{pop}}^{I \rightarrow Y}$ is formulated as:

$$\begin{aligned} w_{G_I G'_I}^{I \rightarrow Y} &= \sum_{i \in \mathcal{Y}} \mathbb{E}_{G_I \sim \mathbb{P}_{G_I}} \mathbb{E}_{G'_I \sim \mathbb{P}_{G'_I}} \mathcal{T}_I(G_I | G_I) \mathcal{T}_I(G'_I | G'_I) \\ &\quad + \mathbb{E}_{G_u \sim \mathbb{P}_u} \mathcal{T}_I(G_I | G_u) \mathcal{T}_I(G'_I | G_u), \end{aligned} \quad (6)$$

where $\mathbb{P}_{G_I}$ is the distribution of samples with class label $i \in \mathcal{Y}$ and $\mathcal{T}_I(G_I | G)$ is the probability that G_I is the invariant subgraph of G .

Then we calculate the normalized adjacency matrix $\bar{\mathbf{A}}_{\text{pop}}^{I \rightarrow Y} = (\mathbf{D}_{\text{pop}}^{I \rightarrow Y})^{-\frac{1}{2}} \mathbf{A}_{\text{pop}}^{I \rightarrow Y} (\mathbf{D}_{\text{pop}}^{I \rightarrow Y})^{-\frac{1}{2}}$, where $\mathbf{A}_{\text{pop}}^{I \rightarrow Y}$ is the adjacency matrix of $G_{\text{pop}}^{I \rightarrow Y}$ and $\mathbf{D}_{\text{pop}}^{I \rightarrow Y}$ is the diagonal degree matrix with $\mathbf{D}_{\text{pop}, G_I G_I}^{I \rightarrow Y} = w_{G_I G_I}^{I \rightarrow Y} = \sum_{G'_I \in \mathcal{G}_I} w_{G_I G'_I}^{I \rightarrow Y}$. Next, we perform a spectral decomposition [23] on $\bar{\mathbf{A}}_{\text{pop}}^{I \rightarrow Y}$ to form a matrix $\mathbf{F}^{I \rightarrow Y^*}$ where the top- k eigenvectors are the columns, and each row can be viewed as a k -dimensional representation of an invariant subgraph [34]. These derived feature representations capture the relational patterns of invariant subgraphs within the given class space $\mathcal{Y}$, thus facilitating generalization to covariate-shifted graphs and enabling discrimination between in-distribution graphs and those with semantic shifts. Specifically, we consider the following optimization, which performs a low-rank matrix approximation on the adjacency matrix:

$$\min_{\mathbf{F}^{I \rightarrow Y}} \mathcal{L}_{\text{mf}}^{I \rightarrow Y} = \|\bar{\mathbf{A}}^{I \rightarrow Y} - \mathbf{F}^{I \rightarrow Y} (\mathbf{F}^{I \rightarrow Y})^\top\|_F^2, \quad (7)$$

where $\|\cdot\|_F$ denotes the matrix Frobenius norm. According to the Eckart-Young-Mirsky theorem [7], the optimal solution of this

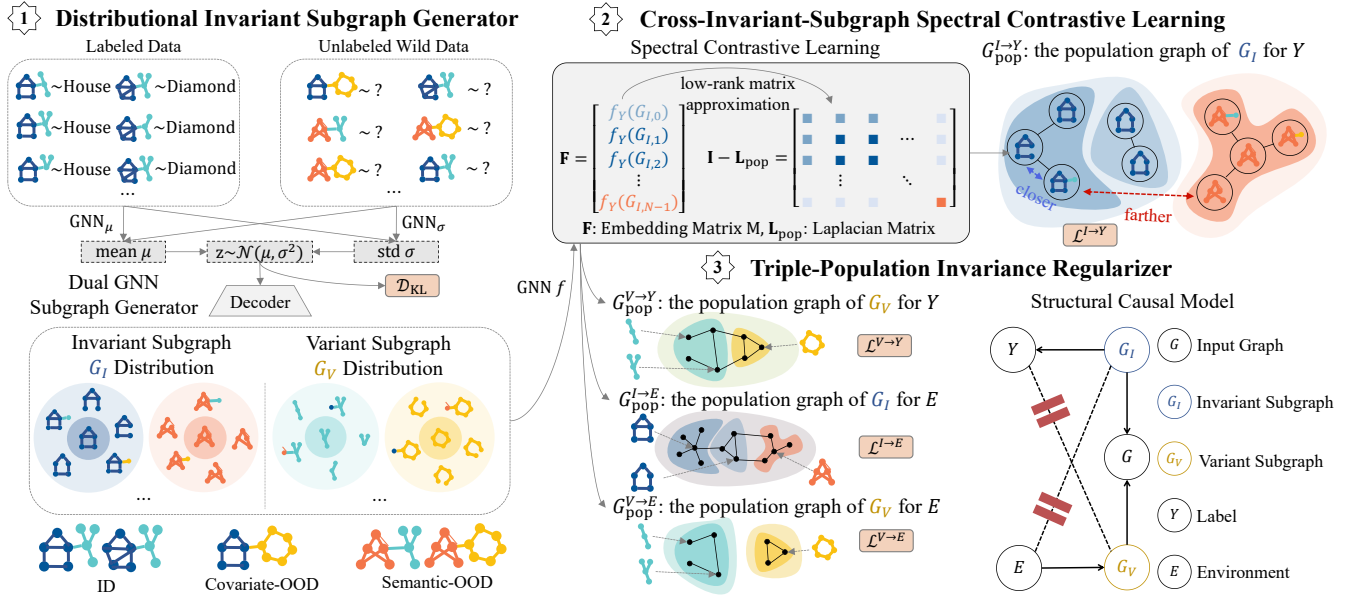


Figure 1: The framework of UniGOOD. Our proposed method jointly optimizes three modules: (1) The distributional invariant subgraph generator utilizes a VGAE-based architecture to model invariant subgraph conditional distributions. (2) The cross-invariant-subgraph spectral contrastive learning module leverages these distributions to learn invariant representations. (3) The triple-population invariance regularizer enforces an invariance principle that identifies invariant subgraphs with stable label relationships through spectral graph theory.

loss function is $\mathbf{F}^{I \rightarrow Y^*}$ such that $\mathbf{F}^{I \rightarrow Y^*} (\mathbf{F}^{I \rightarrow Y^*})^\top$ contains the top- k components of $\mathbf{A}^{I \rightarrow Y}$ eigen decomposition.

In practice, the embedding matrix $\mathbf{F}^{I \rightarrow Y^*}$ cannot be learned with a realistic amount of data. Therefore, we propose to parameterize the rows of $\mathbf{F}^{I \rightarrow Y^*}$ using another GNN $f_Y(\cdot)$. Under this parameterization, we can prove that Eq. (7) becomes equivalent to minimizing the following contrastive learning objective:

$$\mathcal{L}^{I \rightarrow Y} = -2\mathcal{L}_1^{I \rightarrow Y} - 2\mathcal{L}_2^{I \rightarrow Y} + \mathcal{L}_3^{I \rightarrow Y} + 2\mathcal{L}_4^{I \rightarrow Y} + \mathcal{L}_5^{I \rightarrow Y}, \quad (8)$$

where

$$\begin{aligned} \mathcal{L}_1^{I \rightarrow Y} &= \sum_{i \in \mathcal{Y}} \mathbb{E}_{G_I \sim \mathbb{P}_{I_i}, G_I' \sim \mathbb{P}_{I_i}} [f_Y(G_I)^\top f_Y(G_I')], \\ \mathcal{L}_2^{I \rightarrow Y} &= \mathbb{E}_{G_I \sim \mathcal{T}_I(\cdot | G_u), G_I' \sim \mathcal{T}_I(\cdot | G_u)} [f_Y(G_I)^\top f_Y(G_I')], \\ \mathcal{L}_3^{I \rightarrow Y} &= \sum_{i, j \in \mathcal{Y}} \mathbb{E}_{G_I \sim \mathbb{P}_{I_i}, G_I' \sim \mathbb{P}_{I_j}} [(f_Y(G_I)^\top f_Y(G_I'))^2], \\ \mathcal{L}_4^{I \rightarrow Y} &= \sum_{i \in \mathcal{Y}} \mathbb{E}_{G_I \sim \mathbb{P}_{I_i}, G_u \sim \mathbb{P}_u} [(f_Y(G_I)^\top f_Y(G_u^-))^2], \\ \mathcal{L}_5^{I \rightarrow Y} &= \mathbb{E}_{G_u \sim \mathbb{P}_u, G_u' \sim \mathbb{P}_u} [(f_Y(G_u)^\top f_Y(G_u'))^2], \end{aligned} \quad (9)$$

where G_I^+ and G_I^- are the positive and negative invariant subgraphs with respect to G_I , respectively. The proof of equivalence is provided in Appendix A.1. Intuitively, the loss components $\mathcal{L}_1^{I \rightarrow Y}$ and $\mathcal{L}_2^{I \rightarrow Y}$ enhance OOD generalization by pulling positive embedding pairs closer, while $\mathcal{L}_3^{I \rightarrow Y}$, $\mathcal{L}_4^{I \rightarrow Y}$, and $\mathcal{L}_5^{I \rightarrow Y}$ improve OOD detection by pushing negative pairs farther apart. The proof of OOD generalization and detection given invariant subgraphs is provided in Appendix A.2.

3.3 Triple-Population Invariance Regularizer

In this section, with the population graphs G_{pop} of subgraphs, we introduce the invariance principle to identify the invariant subgraphs G_I whose relationship with the label Y is stable for both labeled and unlabeled wild graphs.

To characterize the relationships between the four key elements in invariance theory (invariant subgraphs G_I , variant subgraphs G_V , semantic labels Y , and environment labels E), we formally define three population graphs similar to the framework established by the invariant subgraph population graph $G_{\text{pop}}^{I \rightarrow Y}$ in Section 3.2:

- The population graph of variant subgraphs G_V for the labels Y : $G_{\text{pop}}^{V \rightarrow Y} = (G_V, w^{V \rightarrow Y})$,
- The population graph of invariant subgraphs G_I for the environment labels E : $G_{\text{pop}}^{I \rightarrow E} = (G_I, w^{I \rightarrow E})$,
- The population graph of variant subgraphs G_V for the environment labels E : $G_{\text{pop}}^{V \rightarrow E} = (G_V, w^{V \rightarrow E})$.

G_V is the node set that comprises all possible variant subgraphs extracted from both labeled ID graphs $\mathcal{D}_l$ and unlabeled wild graphs $\mathcal{D}_u$, $w^{V \rightarrow Y}$, $w^{I \rightarrow E}$ and $w^{V \rightarrow E}$ are the edge weights of three population graphs, respectively.

Next, we will elaborate the meaning and calculation of three population graphs. For $G_{\text{pop}}^{V \rightarrow Y}$, since all variant subgraphs exhibit no correlation with the label distribution, they may be considered as an independent class separate from $\mathcal{Y}$. Therefore, we define the edge connectivity of $G_{\text{pop}}^{V \rightarrow Y}$ as follows:

Definition 3. The overall similarity for any pair of data (G_V, G_V') in $G_{\text{pop}}^{V \rightarrow Y}$ is formulated as:

$$w_{G_V, G_V'}^{V \rightarrow Y} = \mathbb{E}_{G_I \sim \mathbb{P}} \mathbb{E}_{G_I' \sim \mathbb{P}} \mathcal{T}_V(G_V | G_I) \mathcal{T}_V(G_V' | G_I'), \quad (10)$$

where $\mathbb{P}$ is the marginal distribution of both labeled/unlabeled data, $\mathcal{T}_V(G_V|G)$ is the probability of G_V being the variant subgraph of G .

For $G_{\text{pop}}^{I \rightarrow E}$, given that all invariant subgraphs exhibit no statistical dependence on environment label distributions (analogous to $G_{\text{pop}}^{V \rightarrow Y}$), we specify the edge connectivity as follows:

Definition 4. The overall similarity for any pair of data (G_I, G'_I) in $G_{\text{pop}}^{I \rightarrow E}$ is formulated as:

$$w_{G_I G'_I}^{I \rightarrow E} = \mathbb{E}_{G_I \sim \mathbb{P}} \mathbb{E}_{G'_I \sim \mathbb{P}} \mathcal{T}_I(G_I|G_I) \mathcal{T}_I(G'_I|G'_I). \quad (11)$$

For $G_{\text{pop}}^{V \rightarrow E}$, given that the environment labels are unknown, we treat all variant subgraphs as unlabeled data. Consequently, the edge connectivity is formally defined as follows:

Definition 5. The overall similarity for any pair of data (G_V, G'_V) in $G_{\text{pop}}^{V \rightarrow E}$ is formulated as:

$$w_{G_V G'_V}^{V \rightarrow E} = \mathbb{E}_{G_u \sim \mathbb{P}} \mathcal{T}_V(G_V|G_u) \mathcal{T}_V(G'_V|G_u). \quad (12)$$

With these definitions and building upon the formulation of $\mathcal{L}^{I \rightarrow Y}$ in Eq. (8), we optimize the following triple-population contrastive learning objectives to derive graph representations:

$$\begin{aligned} \mathcal{L}^{V \rightarrow Y} &= -2 \mathbb{E}_{\substack{G_I \sim \mathbb{P}, G'_I \sim \mathbb{P}, \\ G_V \sim \hat{\mathcal{T}}_V(\cdot|G_I), G'_V \sim \hat{\mathcal{T}}_V(\cdot|G'_I)}} [f_Y(G_V)^\top f_Y(G'_V)], \\ \mathcal{L}^{I \rightarrow E} &= -2 \mathbb{E}_{\substack{G_I \sim \mathbb{P}, G'_I \sim \mathbb{P}, \\ G_I \sim \hat{\mathcal{T}}_I(\cdot|G_I), G'_I \sim \hat{\mathcal{T}}_I(\cdot|G'_I)}} [f_E(G_I)^\top f_E(G'_I)], \\ \mathcal{L}^{V \rightarrow E} &= -2 \mathbb{E}_{\substack{G_u \sim \mathbb{P}, \\ G_V \sim \hat{\mathcal{T}}_V(\cdot|G_u), G'_V \sim \hat{\mathcal{T}}_V(\cdot|G_u)}} [f_E(G_V)^\top f_E(G'_V)] \\ &\quad + \mathbb{E}_{\substack{G_u \sim \mathbb{P}, G'_u \sim \mathbb{P}, \\ G_V \sim \hat{\mathcal{T}}_V(\cdot|G_u), G'_V \sim \hat{\mathcal{T}}_V(\cdot|G'_u)}} [(f_E(G_V)^\top f_E(G'_V))^2], \end{aligned} \quad (13)$$

where $f_E(\cdot)$ denotes another GNN for extracting graph representations to predict environment labels. The invariant subgraph distribution $\hat{\mathcal{T}}_I(\cdot|G)$ is realized by the invariant subgraph generator $\Phi(\cdot)$, such that $\Phi(G) = \hat{\mathcal{T}}_I(\cdot|G)$. Meanwhile, the variant subgraph distribution $\hat{\mathcal{T}}_V(\cdot|G)$ is given by the complement $\hat{\mathcal{T}}_V(\cdot|G) = G \setminus \Phi(G)$.

With four contrastive learning objectives of population graphs, we can prove that our method is theoretically grounded. Formally, we formulate the following invariant subgraph discovery guarantee:

Theorem 1. There exists an optimal invariant subgraph generator $\Phi^*(G) = \mathcal{T}_I(\cdot|G)$ and its complement $\Psi^*(G) = G \setminus \Phi^*(G) = \mathcal{T}_V(\cdot|G)$ if and only if it simultaneously minimizes $\mathcal{L}^{I \rightarrow Y}$, $\mathcal{L}^{V \rightarrow Y}$, $\mathcal{L}^{I \rightarrow E}$, and $\mathcal{L}^{V \rightarrow E}$.

The proof is provided in Appendix A.3. The theorem shows that these four contrastive learning objectives can be jointly employed to simultaneously optimize $\Phi(\cdot)$, $f_Y(\cdot)$, and $f_E(\cdot)$, thereby enabling accurate identification of invariant subgraphs. Our theoretical analysis in Theorem 6 reveals an finding: the population graphs $G_{\text{pop}}^{I \rightarrow Y}$ and $G_{\text{pop}}^{V \rightarrow Y}$ correspond to $G_V \perp Y$, while $G_{\text{pop}}^{I \rightarrow E}$ and $G_{\text{pop}}^{V \rightarrow E}$ correspond to $G_I \perp E$. This shows that **UniGOOD** guarantees the invariance principle through two independence properties [12].

Incorporating $\mathcal{L}^{I \rightarrow Y}$, $\mathcal{L}^{V \rightarrow Y}$, $\mathcal{L}^{I \rightarrow E}$, $\mathcal{L}^{V \rightarrow E}$, and $\mathcal{D}_{\text{KL}}$, the final optimization objective is defined as follows:

$$\min_{\Phi, f_Y, f_E} \mathcal{L}^{I \rightarrow Y} + \beta_1 \mathcal{L}^{V \rightarrow Y} + \beta_2 \mathcal{L}^{I \rightarrow E} + \beta_3 \mathcal{L}^{V \rightarrow E} + \beta_{\text{KL}} \mathcal{D}_{\text{KL}}, \quad (14)$$

where $\beta_1, \beta_2, \beta_3$ and β_{KL} are hyper-parameters to control the impacts of different components. We train a linear or MLP predictor on the invariant subgraph representations $f_Y(\Phi(G))$ for graph label prediction. During inference, the predictor operates on $f_Y(\Phi(G))$ for OOD generalization, while OOD detection leverages the same representations with a non-parametric KNN distance metric that makes no assumptions about the feature space distribution [29].

4 Experiments

We conduct extensive experiments to evaluate **UniGOOD**, focusing on four questions: **RQ1**: Can **UniGOOD** effectively improve OOD detection and generalization abilities? **RQ2**: Can **UniGOOD** identify invariant subgraphs? **RQ3**: Is **UniGOOD** capable of accurately embedding data from varying distributions? **RQ4**: How do the key components and hyper-parameters influence the results?

4.1 Experimental Setup

Datasets. We adopt a synthetic dataset (GD-Motif) with controllable environments and three real-world datasets (GD-HIV-ZINC, GD-Tox21-SIDER, GD-BBBP-BACE) for graph classification.

- **Synthetic Dataset:** Following [39, 43], we construct a synthetic dataset where each graph consists of a variant subgraph and an invariant subgraph, with labels determined solely by the invariant components. The variant subgraphs include Wheel, Tree, Ladder, Star, and Path, while the invariant subgraphs consist of House, Dircycle, Crane, Diamond, Varcycle, and Cross Grid. We generate three distinct distributions: $\mathbb{P}_I$ uses Wheel, Tree, and Ladder as variant subgraphs and House, Dircycle, and Crane as invariant subgraphs; $\mathbb{P}_{\text{co}}$ employs Star and Path as variant subgraphs while keeping the same invariant subgraphs as $\mathbb{P}_I$; and $\mathbb{P}_{\text{se}}$ includes all variant subgraphs paired with Diamond, Varcycle, and Cross Grid as invariant subgraphs.
- **Real-world Dataset:** We apply the scaffold split of one dataset to generate $\mathbb{P}_I$ and $\mathbb{P}_{\text{co}}$ and another dataset to generate $\mathbb{P}_{\text{se}}$. For example, GD-HIV-ZINC applies the scaffold split of HIV [40] to generate $\mathbb{P}_I$ and $\mathbb{P}_{\text{co}}$, and applies ZINC [10] to generate $\mathbb{P}_{\text{se}}$.

To simulate the wild distribution $\mathbb{P}_u$, we adopt the same mixture ratio as in Scone [2] and Bridge [34], where $\pi_c = 0.5$ and $\pi_s = 0.1$.

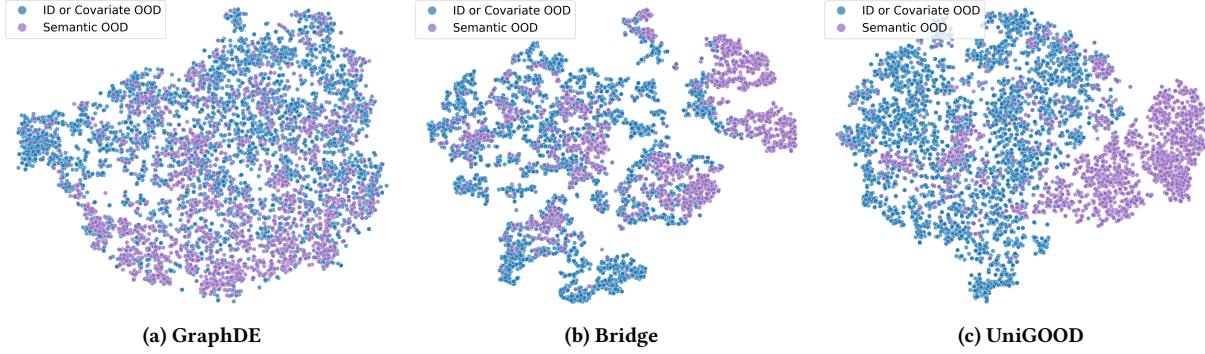
Baselines. We compare **UniGOOD** with state-of-the-art methods from three groups. The first group consists of graph OOD detection methods, including Energy [21], GraphDE [19], GNNSafe [37], AAGOD [13] and GOODAT [36]. The second group is graph OOD generalization methods that include standard ERM, IRM [1], DIR [39], GRE [20], GIL [17], GSAT [22], CIGA [6], iMoLD [48], and CAL+ [28]. The third group includes methods that simultaneously address OOD generalization and detection, including Scone [2] and Bridge [34].

Metrics. For ID generalization performance, we use ACC for $\mathbb{P}_I$ in GD-Motif and AUROC for other three datasets. For OOD generalization performance, we use ACC for $\mathbb{P}_{\text{co}}$ in GD-Motif and AUROC for other three datasets. For OOD detection performance, ID graphs from $\mathbb{P}_I$ are considered positive, and semantic-shifted OOD graphs from $\mathbb{P}_{\text{se}}$ are considered negative. We use two widely used metrics [2, 13, 19, 34]: FPR95 and AUROC.

Implementation Details. We employ a 3-layer GIN with 256-dimensional hidden representations and virtual nodes. The hyper-parameters β_1, β_2 , and β_3 are set at 1, while β_{KL} is chosen from

Table 1: Performance comparison with competitive graph OOD generalization and detection methods on GD-Motif and GD-HIV-ZINC. Numbers after $\pm$ indicate variances. The best results are in bold and the second-best results are underlined.

	GD-Motif				GD-HIV-ZINC			
	Generalization		Detection		Generalization		Detection	
	ID $\uparrow$	OOD $\uparrow$	FPR $\downarrow$	AUROC $\uparrow$	ID $\uparrow$	OOD $\uparrow$	FPR $\downarrow$	AUROC $\uparrow$
<i>OOD Detection</i>								
Energy	53.17 $\pm$ 8.00	45.40 $\pm$ 8.94	89.32 $\pm$ 8.20	48.25 $\pm$ 1.79	74.13 $\pm$ 3.54	61.28 $\pm$ 6.82	99.86 $\pm$ 0.17	28.51 $\pm$ 1.33
GraphDE	93.25 $\pm$ 0.17	57.66 $\pm$ 4.56	98.11 $\pm$ 0.00	42.28 $\pm$ 0.00	82.09 $\pm$ 1.22	72.13 $\pm$ 1.08	87.00 $\pm$ 7.85	67.00 $\pm$ 4.74
GNNSafe	92.88 $\pm$ 0.05	61.40 $\pm$ 0.78	49.62 $\pm$ 10.63	79.49 $\pm$ 6.53	80.42 $\pm$ 0.31	68.88 $\pm$ 2.76	97.44 $\pm$ 0.76	36.80 $\pm$ 4.58
AAGOD	92.42 $\pm$ 0.15	61.58 $\pm$ 0.81	16.18 $\pm$ 4.79	97.16 $\pm$ 1.22	81.43 $\pm$ 1.39	66.46 $\pm$ 0.31	99.59 $\pm$ 0.01	45.60 $\pm$ 2.68
GOODAT	92.52 $\pm$ 0.22	59.70 $\pm$ 1.21	96.44 $\pm$ 1.83	57.38 $\pm$ 7.45	82.66 $\pm$ 1.29	71.55 $\pm$ 1.04	99.99 $\pm$ 0.02	36.30 $\pm$ 9.80
<i>OOD Generalization</i>								
ERM	92.46 $\pm$ 0.11	60.30 $\pm$ 2.12	13.27 $\pm$ 5.49	97.77 $\pm$ 0.69	82.19 $\pm$ 0.30	68.57 $\pm$ 1.97	97.90 $\pm$ 0.90	38.74 $\pm$ 0.62
IRM	92.33 $\pm$ 0.15	59.49 $\pm$ 7.90	13.18 $\pm$ 3.67	97.71 $\pm$ 0.60	81.12 $\pm$ 1.42	68.26 $\pm$ 0.53	96.11 $\pm$ 3.65	43.23 $\pm$ 6.95
DIR	82.68 $\pm$ 2.71	55.01 $\pm$ 11.29	23.38 $\pm$ 1.47	95.12 $\pm$ 0.33	80.36 $\pm$ 2.33	68.02 $\pm$ 0.68	98.02 $\pm$ 0.36	44.62 $\pm$ 3.89
GREa	<u>93.32</u> $\pm$ 0.02	64.00 $\pm$ 4.34	14.01 $\pm$ 3.64	97.51 $\pm$ 0.84	77.00 $\pm$ 1.91	<u>72.84</u> $\pm$ 0.66	98.75 $\pm$ 1.08	51.46 $\pm$ 10.41
GIL	87.56 $\pm$ 3.06	64.83 $\pm$ 4.78	58.69 $\pm$ 9.48	86.89 $\pm$ 3.12	77.29 $\pm$ 1.21	65.05 $\pm$ 5.37	98.98 $\pm$ 1.41	47.05 $\pm$ 6.43
GSAT	77.17 $\pm$ 7.64	56.27 $\pm$ 1.21	26.19 $\pm$ 2.66	95.07 $\pm$ 0.81	81.53 $\pm$ 2.06	68.90 $\pm$ 0.98	97.24 $\pm$ 1.67	43.57 $\pm$ 3.71
CIGA	91.98 $\pm$ 0.76	66.36 $\pm$ 10.95	24.36 $\pm$ 10.64	93.67 $\pm$ 3.74	80.10 $\pm$ 1.55	68.92 $\pm$ 0.63	96.26 $\pm$ 1.43	42.38 $\pm$ 4.44
iMoLD	92.58 $\pm$ 0.23	61.68 $\pm$ 7.25	15.62 $\pm$ 2.08	97.38 $\pm$ 0.22	81.64 $\pm$ 0.80	68.82 $\pm$ 1.82	97.60 $\pm$ 1.06	46.03 $\pm$ 2.56
CAL+	93.39 $\pm$ 0.19	<u>68.02</u> $\pm$ 1.22	20.82 $\pm$ 1.58	95.02 $\pm$ 1.02	80.24 $\pm$ 0.64	71.86 $\pm$ 0.33	98.33 $\pm$ 0.45	32.74 $\pm$ 1.33
<i>Joint OOD Generalization and Detection</i>								
Scone	92.42 $\pm$ 0.31	59.17 $\pm$ 11.68	77.57 $\pm$ 5.94	54.08 $\pm$ 19.53	77.11 $\pm$ 4.35	68.85 $\pm$ 7.14	99.47 $\pm$ 0.47	33.03 $\pm$ 8.57
Bridge	87.73 $\pm$ 1.38	39.57 $\pm$ 1.60	<u>13.06</u> $\pm$ 2.18	<u>98.10</u> $\pm$ 0.39	80.10 $\pm$ 1.24	57.40 $\pm$ 3.57	<u>58.59</u> $\pm$ 7.55	<u>80.18</u> $\pm$ 2.14
UniGOOD	88.32 $\pm$ 0.46	74.01 $\pm$ 6.17	8.14 $\pm$ 4.86	98.39 $\pm$ 1.04	79.87 $\pm$ 1.21	73.87 $\pm$ 3.07	53.19 $\pm$ 11.70	81.49 $\pm$ 4.27

**Figure 2: The t-SNE visualization of the learned invariant graph representations on GD-HIV-ZINC.**

{10, 20}. Model selection used both ID and wild validation loss, while fine-tuning and hyperparameter tuning relied solely on ID validation performance. All experimental results are reported as the mean $\pm$ standard deviation across 3 independent runs.

4.2 Performance Comparisons

To answer **RQ1**, we report the results of **UniGOOD** with baselines across all datasets in Tables 1 and 2. Our method achieves the best performance for all synthetic and real-world datasets, excelling in both OOD generalization and detection tasks.

On the synthetic GD-Motif dataset, which serves as a crucial testing ground for both the generalization to covariate shifts and the

detection to semantic shifts, **UniGOOD** achieves state-of-the-art performance. For OOD generalization, it shows an 8.81% improvement in accuracy relative to the second-best method. Meanwhile, it reduces FPR by 37.67% compared to competing approaches. The results clearly verify the ability of our method to reliably generalize to covariate-shifted graphs and detect semantic-shifted graphs.

The advantages of **UniGOOD** are also shown in real-world datasets. Our method demonstrates superior OOD generalization, achieving relative gains of 9.53% in AUROC scores compared to baselines on GD-Tox21-SIDER. Beyond generalization, **UniGOOD** also excels in OOD detection, with a 9.21% relative reduction in FPR on GD-HIV-ZINC. These results highlight its effectiveness on graph OOD generalization and detection.

Table 2: Performance comparison with competitive graph OOD generalization and detection methods on GD-Tox21-SIDER and GD-BBBP-BACE. Numbers after $\pm$ indicate variances. The best results are in bold and the second-best results are underlined.

	GD-Tox21-SIDER				GD-BBBP-BACE			
	Generalization		Detection		Generalization		Detection	
	ID $\uparrow$	OOD $\uparrow$	FPR $\downarrow$	AUROC $\uparrow$	ID $\uparrow$	OOD $\uparrow$	FPR $\downarrow$	AUROC $\uparrow$
<i>OOD Detection</i>								
Energy	77.58 $\pm$ 4.48	69.89 $\pm$ 2.14	89.29 $\pm$ 2.35	<u>68.96</u> $\pm$ 3.46	79.34 $\pm$ 5.12	64.79 $\pm$ 14.65	98.35 $\pm$ 2.01	63.52 $\pm$ 9.97
GraphDE	80.65 $\pm$ 2.37	63.01 $\pm$ 5.93	98.45 $\pm$ 0.21	56.24 $\pm$ 0.26	88.00 $\pm$ 2.50	73.01 $\pm$ 3.85	95.27 $\pm$ 0.19	61.23 $\pm$ 0.65
GNNSafe	84.12 $\pm$ 0.65	59.64 $\pm$ 5.43	92.26 $\pm$ 4.47	57.79 $\pm$ 12.00	83.32 $\pm$ 0.44	72.34 $\pm$ 3.76	94.28 $\pm$ 5.01	54.17 $\pm$ 9.17
AAGOD	81.78 $\pm$ 2.33	61.11 $\pm$ 7.13	87.74 $\pm$ 0.90	57.69 $\pm$ 5.89	88.94 $\pm$ 3.97	80.48 $\pm$ 3.83	86.47 $\pm$ 1.40	<u>72.92</u> $\pm$ 2.72
GOODAT	78.99 $\pm$ 1.18	65.70 $\pm$ 2.43	<u>86.43</u> $\pm$ 2.84	63.68 $\pm$ 5.48	84.17 $\pm$ 2.74	74.54 $\pm$ 6.57	89.22 $\pm$ 9.58	61.23 $\pm$ 13.83
<i>OOD Generalization</i>								
ERM	82.28 $\pm$ 2.79	59.31 $\pm$ 12.78	87.98 $\pm$ 1.48	60.16 $\pm$ 7.81	86.82 $\pm$ 1.61	80.92 $\pm$ 3.47	96.59 $\pm$ 2.20	57.45 $\pm$ 4.29
IRM	81.55 $\pm$ 1.83	60.39 $\pm$ 7.78	87.02 $\pm$ 1.15	59.72 $\pm$ 4.85	89.38 $\pm$ 1.89	80.92 $\pm$ 4.45	96.37 $\pm$ 0.87	54.34 $\pm$ 2.71
DIR	77.05 $\pm$ 3.56	66.03 $\pm$ 1.41	88.69 $\pm$ 7.05	53.95 $\pm$ 14.38	88.43 $\pm$ 0.71	<u>82.06</u> $\pm$ 5.59	91.09 $\pm$ 5.43	62.56 $\pm$ 8.45
GREa	<u>83.94</u> $\pm$ 0.72	68.03 $\pm$ 4.83	90.24 $\pm$ 1.44	56.59 $\pm$ 3.16	83.56 $\pm$ 1.76	78.30 $\pm$ 0.11	94.06 $\pm$ 3.45	55.70 $\pm$ 4.40
GIL	74.83 $\pm$ 6.82	65.26 $\pm$ 7.74	89.64 $\pm$ 0.72	62.71 $\pm$ 10.46	83.82 $\pm$ 2.31	69.97 $\pm$ 5.70	95.82 $\pm$ 4.67	56.61 $\pm$ 10.30
GSAT	81.20 $\pm$ 3.28	64.38 $\pm$ 4.45	87.74 $\pm$ 0.55	67.23 $\pm$ 2.33	87.85 $\pm$ 3.13	78.90 $\pm$ 5.07	95.27 $\pm$ 3.94	58.84 $\pm$ 4.25
CIGA	78.85 $\pm$ 3.20	70.19 $\pm$ 2.14	92.98 $\pm$ 3.38	56.02 $\pm$ 16.47	86.48 $\pm$ 0.34	78.47 $\pm$ 1.85	92.08 $\pm$ 6.86	62.30 $\pm$ 10.44
iMoLD	80.74 $\pm$ 3.60	63.95 $\pm$ 6.43	87.98 $\pm$ 0.20	62.05 $\pm$ 2.93	83.67 $\pm$ 1.51	77.27 $\pm$ 2.61	97.80 $\pm$ 1.82	53.59 $\pm$ 17.59
CAL+	81.00 $\pm$ 1.32	<u>73.14</u> $\pm$ 5.05	92.62 $\pm$ 1.61	56.42 $\pm$ 3.66	<u>88.98</u> $\pm$ 1.79	81.42 $\pm$ 1.81	92.30 $\pm$ 4.20	62.82 $\pm$ 5.31
<i>Joint OOD Generalization and Detection</i>								
Scone	81.61 $\pm$ 2.20	63.10 $\pm$ 7.09	88.81 $\pm$ 2.30	66.57 $\pm$ 5.06	85.75 $\pm$ 2.14	74.79 $\pm$ 3.96	96.81 $\pm$ 2.20	56.16 $\pm$ 12.06
Bridge	80.48 $\pm$ 2.43	60.88 $\pm$ 9.99	87.38 $\pm$ 4.18	66.72 $\pm$ 2.30	75.10 $\pm$ 2.33	71.42 $\pm$ 8.75	80.53 $\pm$ 3.02	70.51 $\pm$ 4.29
UniGOOD	80.13 $\pm$ 1.14	80.11 $\pm$ 1.82	83.93 $\pm$ 3.52	69.98 $\pm$ 0.34	84.55 $\pm$ 1.31	82.70 $\pm$ 1.24	77.89 $\pm$ 16.21	73.29 $\pm$ 4.92

Table 3: The precision@10 of discovering the ground-truth invariant subgraphs on GD-Motif.

	Covariate OOD	Semantic OOD
DIR	0.368 $\pm$ 0.236	0.274 $\pm$ 0.085
GIL	0.075 $\pm$ 0.030	0.237 $\pm$ 0.016
GSAT	0.407 $\pm$ 0.122	0.386 $\pm$ 0.093
CIGA	0.340 $\pm$ 0.004	0.310 $\pm$ 0.032
UniGOOD	0.423 $\pm$ 0.115	0.485 $\pm$ 0.127

4.3 In-Depth Analyses

4.3.1 Invariant Subgraphs. For **RQ2**, we evaluate our method against baselines which also explicitly generate subgraphs via edge masking. Specifically, we evaluate the identified invariant subgraphs against ground-truth on GD-Motif under covariate and semantic shifts, using Precision@10. Results in Table 3 show that **UniGOOD** outperforms baseline methods in discovering the underlying invariant subgraphs, particularly under semantic shifts.

4.3.2 Visualization. For **RQ3**, we visualize the learned invariant graph representations using t-SNE [32] and compare with top-performing baselines, GraphDE and Bridge. As shown in Figure 2, our framework achieves relatively clear separation between semantic-shifted OOD graphs and both ID/covariate-shifted graphs in the embedding space, while the baseline methods fail to maintain such clear separation boundaries. This discriminative capability of our approach significantly enhances OOD detection performance.

Table 4: Ablation studies on UniGOOD using GD-Motif, with the optimal performance highlighted in bold.

	ID Acc. $\uparrow$	OOD Acc. $\uparrow$	FPR $\downarrow$	AUROC $\uparrow$
UniGOOD	88.32 $\pm$ 0.46	74.01 $\pm$ 6.17	8.14 $\pm$ 4.86	98.39 $\pm$ 1.04
w/o $\mathcal{D}_{KL}$	70.73 $\pm$ 8.95	66.02 $\pm$ 8.14	21.31 $\pm$ 8.75	94.90 $\pm$ 2.35
w/o $\mathcal{L}^{V \rightarrow Y}$	85.48 $\pm$ 1.22	69.28 $\pm$ 5.18	12.60 $\pm$ 7.44	97.42 $\pm$ 1.64
w/o $\mathcal{L}^{I \rightarrow E}$	85.96 $\pm$ 3.22	71.64 $\pm$ 11.29	14.48 $\pm$ 7.31	96.44 $\pm$ 1.78
w/o $\mathcal{L}^{V \rightarrow E}$	86.63 $\pm$ 3.70	68.72 $\pm$ 7.48	8.83 $\pm$ 3.61	97.99 $\pm$ 1.48

4.3.3 Ablation Studies. To answer **RQ4**, we first evaluate two key modules of **UniGOOD**: the distributional invariant subgraph generator and the triple-population invariance regularizer. For the distributional invariant subgraph generator, we compare with removing the KL loss, retaining only the subgraph generator (denoted as w/o $\mathcal{D}_{KL}$). For the triple-population invariance regularizer, we remove each of its triple loss items, denoted as w/o $\mathcal{L}^{V \rightarrow Y}$, w/o $\mathcal{L}^{I \rightarrow E}$, and w/o $\mathcal{L}^{V \rightarrow E}$, respectively. The results on the GD-Motif dataset are shown in Table 4, while other datasets show similar patterns. We have the following observations:

By removing the KL loss, our method achieves suboptimal performance, suggesting that the KL divergence plays a crucial role in generating diverse invariant subgraphs. Without this regularization, the distributional invariant subgraph generator degenerates to a standard GAE, losing its ability to provide informative prior knowledge, particularly for unlabeled data.

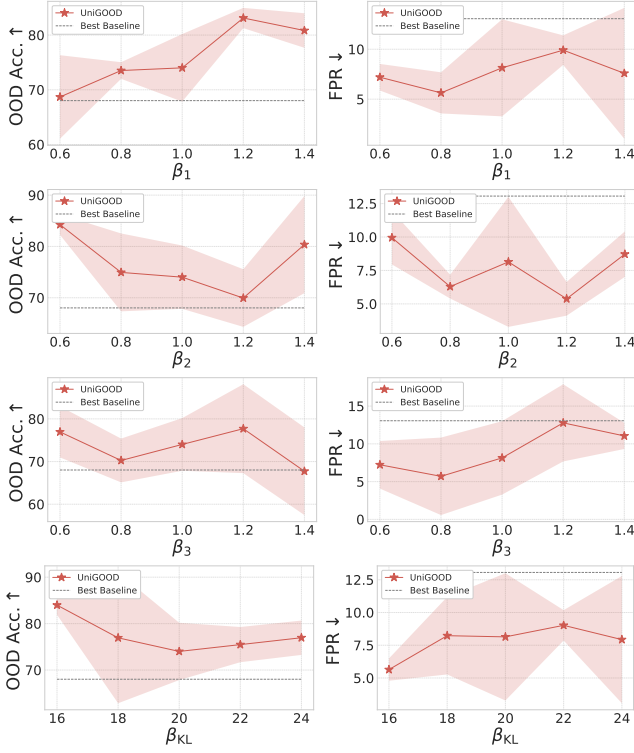


Figure 3: The results of hyper-parameter sensitivity.

Ablation of any individual loss component in the triple-population invariance regularizer similarly leads to performance degradation. The results confirm that our specifically designed invariance principle enhances the identification of invariant subgraphs, thereby improving both OOD generalization and detection performance.

4.3.4 Hyper-parameter Sensitivity. We further evaluate the sensitivity of hyper-parameters, including the contrastive learning loss coefficients β_1 , β_2 , β_3 and the KL loss coefficient β_{KL} . For simplicity, we only report the results on GD-Motif in Figure 3, while results on other datasets show similar patterns. The contrastive learning loss coefficients β_1 , β_2 , and β_3 exhibit moderate influence on model performance. Excessively large values may dominate other loss terms, while overly small values prove inadequate. The KL loss coefficient β_{KL} is a critical parameter. Setting it too high may lead to the model generating unrealistic subgraphs, while setting it too low can result in insufficient diversity in the generated subgraphs. Although an appropriate choice can further improve the performance, our method is not very sensitive to hyper-parameters and outperforms the best baselines with a wide range of hyper-parameters choices.

5 Related Work

Graph Invariant Learning for OOD Generalization. To tackle the distribution shift problem of graph data, graph invariant learning has received widespread attention [16]. These approaches generally consider distribution shifts as stemming from unknown environmental variables [12, 20], focusing on invariant aspects of the data for prediction [16, 39]. For example, GIL [17] captures invariant relationships between graph structure and labels with three

modules: invariant subgraph identification, environment inference, and invariant learning. DIR [38] similarly employs a GNN-based model to split the input graph into invariant and variant components, which are then processed by a distribution intervener to minimize invariant risk across varying environments. Other methods such as EERM [38], DIDA [46], and StableGNN [8] also explore similar concepts. Despite their progress, these methods focus on generalization, but do not consider OOD detection problem.

Graph OOD Detection. Recent advances in Graph OOD Detection have focused on identifying graph data that deviates from the training distribution to enhance model performance in real-world applications. Existing methods can be broadly categorized into four categories [4]: (1) Enhancement-based methods improve OOD sensitivity through data augmentation [36] or model architecture modifications [33]; (2) Reconstruction-based techniques, such as PGR-MOOD [25], leverage generative models like VGAEs or diffusion models to detect OOD samples by measuring reconstruction errors; (3) Information propagation-based methods exploit graph structures to propagate features [26], energy scores [37] or uncertainty estimates [27] for OOD identification; (4) Classification-based approaches, including probabilistic evidence methods [47] and boundary optimization techniques [13], refine decision boundaries to distinguish IN and OOD samples. Despite their advancements, existing methods prioritize high sensitivity for accurate OOD sample identification without considering generalization performance. In our paper, we provide a novel framework to unify graph OOD generalization and detection.

Spectral Contrastive Learning. They have been widely adopted in various machine learning problems, including self-supervised learning [14], unsupervised domain adaptation [24], novel category discovery [31], and open-world semi-supervised learning [30]. Bridge [34] introduces a novel framework with distinct learning objectives, focusing primarily on joint OOD generalization and detection. However, Bridge was originally designed for computer vision tasks with hand-crafted data augmentation strategies, and extending it to graph-structured data faces fundamental limitations. Graph distribution shifts exhibit unique complexities that differ significantly from the Euclidean data. Furthermore, Bridge’s heavy reliance on data augmentation poses additional challenges for graphs, where effective augmentation needs to consider the inherent irregularity and topological complexity of real-world graph data. In experiments, our method also greatly outperforms Bridge.

6 Conclusion

In this work, we propose **UniGOOD**, a novel framework for joint graph OOD generalization and detection. By integrating three key components, including the distributional invariant subgraph generator, the cross-invariant-subgraph spectral contrastive learning module, and the triple-population invariance regularizer, our method effectively handles both tasks. Theoretical analysis and extensive experiments validate the superiority of **UniGOOD**.

Acknowledgments

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A Theory

Definition 6. Let $G_{\text{pop}}^{I \rightarrow Y} = (\mathcal{G}_I, w_{G_I}^{I \rightarrow Y})$ be the population graph of invariant subgraphs. Its normalized adjacency and Laplacian matrices are defined as $\bar{\mathbf{A}}_{\text{pop}}^{I \rightarrow Y} = (\mathbf{D}_{\text{pop}}^{I \rightarrow Y})^{-\frac{1}{2}} \mathbf{A}_{\text{pop}}^{I \rightarrow Y} (\mathbf{D}_{\text{pop}}^{I \rightarrow Y})^{-\frac{1}{2}}$ and $\mathbf{L}_{\text{pop}}^{I \rightarrow Y} = \mathbf{I} - \bar{\mathbf{A}}_{\text{pop}}^{I \rightarrow Y}$, respectively, where $\mathbf{A}_{\text{pop}}^{I \rightarrow Y}$ has entries $\mathbf{A}_{\text{pop}, G_I G_I'}^{I \rightarrow Y}$ and $\mathbf{D}_{\text{pop}}^{I \rightarrow Y}$ is the diagonal degree matrix with $\mathbf{D}_{\text{pop}}^{I \rightarrow Y} = w_{G_I}$.

A.1 Equivalence between Two Objectives

Theorem 2. Let $f_Y(\cdot)$ be a feature extractor, and let matrix $\mathbf{F}^{I \rightarrow Y}$ be constructed such that each of its rows is given by $\sqrt{w_{G_I}^{I \rightarrow Y}} \cdot f_Y(G_I)$. Then, $\mathbf{F}^{I \rightarrow Y}$ is a minimizer of $\mathcal{L}_{\text{mf}}^{I \rightarrow Y}$ if and only if $f_Y(\cdot)$ is a minimizer of the spectral contrastive loss $\mathcal{L}^{I \rightarrow Y}$.

PROOF. We begin by expanding $\mathcal{L}_{\text{mf}}^{I \rightarrow Y}$:

$$\begin{aligned} \mathcal{L}_{\text{mf}}^{I \rightarrow Y} &= \sum_{G_I, G_I' \in \mathcal{G}_I} w_{G_I}^{I \rightarrow Y} w_{G_I'}^{I \rightarrow Y} (f_Y(G_I)^\top f_Y(G_I'))^2 \\ &\quad - 2 \sum_{G_I, G_I' \in \mathcal{G}_I} w_{G_I}^{I \rightarrow Y} w_{G_I'}^{I \rightarrow Y} f_Y(G_I)^\top f_Y(G_I') + c, \end{aligned} \quad (15)$$

where c represents a constant term.

$$w_{G_I}^{I \rightarrow Y} = \sum_{G_I'} w_{G_I G_I'}^{I \rightarrow Y} = \sum_{i \in \mathcal{Y}} \mathbb{E}_{G_I \sim \mathbb{P}_{I_i}} \mathcal{T}_I(G_I | G_I) + \mathbb{E}_{G_u \sim \mathbb{P}_u} \mathcal{T}_I(G_I | G_u) \quad (16)$$

The loss form depends on the definition of positive/negative pairs. Substituting $w_{G_I G_I'}^{I \rightarrow Y}$ yields:

$$\begin{aligned} &- 2 \sum_{G_I, G_I' \in \mathcal{G}_I} w_{G_I G_I'}^{I \rightarrow Y} f_Y(G_I)^\top f_Y(G_I') \\ &= - 2 \sum_{i \in \mathcal{Y}} \mathbb{E}_{G_I \sim \mathbb{P}_{I_i}} \mathbb{E}_{G_I' \sim \mathbb{P}_{I_i}} \sum_{G_I, G_I' \in \mathcal{G}_I} \mathcal{T}_I(G_I | G_I) \mathcal{T}_I(G_I' | G_I') f_Y(G_I)^\top f_Y(G_I') \\ &\quad - 2 \mathbb{E}_{G_u \sim \mathbb{P}_u} \sum_{G_I, G_I' \in \mathcal{G}_I} \mathcal{T}_I(G_I | G_u) \mathcal{T}_I(G_I' | G_u) f_Y(G_I)^\top f_Y(G_I') \\ &= - 2 \mathcal{L}_1^{I \rightarrow Y} - 2 \mathcal{L}_2^{I \rightarrow Y} \end{aligned} \quad (17)$$

Similarly, incorporating both $w_{G_I}^{I \rightarrow Y}$ and $w_{G_I'}^{I \rightarrow Y}$ yields [34]:

$$\sum_{G_I, G_I' \in \mathcal{G}_I} w_{G_I}^{I \rightarrow Y} w_{G_I'}^{I \rightarrow Y} (f_Y(G_I)^\top f_Y(G_I'))^2 = \mathcal{L}_3^{I \rightarrow Y} + 2 \mathcal{L}_4^{I \rightarrow Y} + \mathcal{L}_5^{I \rightarrow Y} \quad (18)$$

Combining these expressions yields the final relationship:

$$\mathcal{L}_{\text{mf}}^{I \rightarrow Y} = -2 \mathcal{L}_1^{I \rightarrow Y} - 2 \mathcal{L}_2^{I \rightarrow Y} + \mathcal{L}_3^{I \rightarrow Y} + 2 \mathcal{L}_4^{I \rightarrow Y} + \mathcal{L}_5^{I \rightarrow Y} + c. \quad (19)$$

Since c is an additive constant, this completes the proof. $\square$

Similar relationships hold for $G^{V \rightarrow Y}$, $G^{I \rightarrow E}$, and $G^{V \rightarrow E}$; we omit the analogous derivations for brevity.

A.2 OOD Generalization and Detection

We assume that each graph $G \in \mathcal{G}$ belongs to one of the $r_Y = r_Y^I + r_Y^u$ classes, with r_Y^I known (in $\mathcal{Y}$) and r_Y^u unknown. An extended label function $\hat{y} : \mathcal{G}_I \rightarrow [r_Y]$ assigns each invariant subgraph the same label as its original graph.

Assumption 1. For $G \sim \mathbb{P}$ and $y(G)$ to be its label. Let the invariant subgraphs $G_I \sim \mathcal{T}_I(\cdot | G)$. We assume that there exists a classifier $g_Y(\cdot)$ that can predict $y(G)$ given G_I with error at most $\alpha^{I \rightarrow Y} = 0$. That is, $g_Y(G_I) = y(G)$.

Given the extended labeled function, we define the following quantity that describes the difference between extended labels of two invariant subgraphs: $\phi^{I \rightarrow Y} = \sum_{G_I, G_I' \in \mathcal{G}_I} w_{G_I, G_I'} \mathbb{1}[\hat{y}(G_I) \neq \hat{y}(G_I')]$. We also define the following three quantities that describe the difference between the extended label and ground truth label:

$$\Delta_I^I(y, \hat{y}) = \Pr_{G_I \sim \mathbb{P}_{I_i}, G_I \sim \mathcal{T}_I(\cdot | G_I)} (\hat{y}(G_I) \neq y(G_I)), \quad (20)$$

$$\Delta_I^I(y, \hat{y}) = \Pr_{G_I \sim \mathbb{P}_{I_i}, G_I \sim \mathcal{T}_I(\cdot | G_I)} (\hat{y}(G_I) \neq y(G_I)), \quad (21)$$

$$\Delta_I^u(y, \hat{y}) = \Pr_{G_u \sim \mathbb{P}_u, G_I \sim \mathcal{T}_I(\cdot | G_u)} (\hat{y}(G_I) \neq y(G_u)), \quad (22)$$

In accordance with Assumption 1, all three quantities evaluate to 0.

Our method utilizes both the labeled set $\mathcal{D}_I$ and unlabeled wild set $\mathcal{D}_u$ to pretrain the feature extractor $f_Y(\cdot)$, while only $\mathcal{D}_I$ is used for downstream classifier training. For generalization error bounds of invariant subgraph classification, we employ the end-to-end bounds from [14] for finite samples.

Theorem 3. Choose $C_d^{I \rightarrow Y} > 0$ such that $C_d^{I \rightarrow Y} \leq \frac{2(d+1)}{Y_d^{I \rightarrow Y}}$, where

$Y_d^{I \rightarrow Y}$ is the d largest eigenvalues of $\bar{\mathbf{A}}_{\text{pop}}^{I \rightarrow Y}$. Then, with probability at least $1 - \delta$ over the randomness of data, for any $f_Y^* \in \mathcal{F}$ that minimizes the contrastive learning loss $\mathcal{L}^{I \rightarrow Y}$ in Eq. (8) and a linear head $\hat{\mathbf{B}}^{I \rightarrow Y} \in \arg \min_{\|\mathbf{B}\|_F \leq C_d^{I \rightarrow Y}} \sum_{i=1}^n \sum_{i=1}^{r_Y^I} \min\{(\mathbf{B}^\top f_Y^*(G_{I,i}) - \tilde{y}(G_{I,i}))^2, 1\}$, where $\tilde{y}(G)$ is the one-hot embedding of $y(G)$, we have:

$$\begin{aligned} \mathcal{E}^{I \rightarrow Y} &\lesssim \frac{cd}{\Delta_Y^{I \rightarrow Y}} \left(\hat{\mathcal{R}}_{(n+m)/2}(\mathcal{F}) + \sqrt{\frac{\log 2/\delta}{n+m}} + \delta \right) \\ &\quad + \left(r_Y^I C_d \sqrt{\frac{d}{n}} + \sqrt{\frac{\log 1/\delta}{n}} \right), \end{aligned} \quad (23)$$

where $\mathcal{E}^{I \rightarrow Y} = \Pr_{G \sim \mathbb{P}, G_I \sim \mathcal{T}_I(\cdot | G)} [y(G) \neq g_{f_Y^*, \hat{\mathbf{B}}^{I \rightarrow Y}}(G_I)]$ denotes the generalization error of the representation $f_Y^*(G)$ and the linear head $\hat{\mathbf{B}}^{I \rightarrow Y}$. $g_{f_Y^*, \hat{\mathbf{B}}^{I \rightarrow Y}}(G) = \arg \max_{i \in [r_Y]} (f_Y^*(G)^\top \hat{\mathbf{B}}^{I \rightarrow Y})_i$ is the graph predictor. c is a constant, and $\Delta_Y^{I \rightarrow Y} = Y_{[3d/4]}^{I \rightarrow Y} - Y_d^{I \rightarrow Y}$ is the eigenvalue gap. The Rademacher complexity $\hat{\mathcal{R}}_{(n+m)/2}(\mathcal{F})$ is roughly $\sqrt{2R/(n+m)}$, with R measures the complexity of $\mathcal{F}$.

Compared to [14], Theorem 3 modifies the notation and, under Assumption 1, simplifies as $\alpha^{I \rightarrow Y}$, Δ_I^I , Δ_I^I , $\Delta_I^u = 0$, thereby proving OOD generalization with invariant subgraphs:

Lemma 1. In the setting of Theorem 3, assume labeled and unlabeled wild sets are large enough, i.e., $m+n = \text{poly}(d, \frac{1}{\Delta_Y^{I \rightarrow Y}}, R)$ and $n = \text{poly}(d, r_Y^I, \frac{1}{Y_d^{I \rightarrow Y}})$, given invariant subgraphs $G_I \sim \mathcal{T}_I(\cdot | G)$. Then, with probability at least $1 - \delta$ over the randomness of data, for any $f_Y^*(\cdot)$ be the optimal GNN minimizing Eq. (8) and a linear head $\hat{\mathbf{B}}^{I \rightarrow Y} \in \arg \min_{\|\mathbf{B}\|_F \leq C_d^{I \rightarrow Y}} \sum_{i=1}^n \sum_{i=1}^{r_Y^I} \min\{(\mathbf{B}^\top f_Y^*(G_{I,i}) - \tilde{y}(G_{I,i}))^2, 1\}$, we have a satisfactory graph predictor for covariate-shifted OOD graphs.

PROOF. Pursuant to Eq. (23), we prove $\mathcal{E}^{I \rightarrow Y} \rightarrow 0$ by analyzing the vanishing criteria for each component, which relies on the polynomial growth of sample sizes (n, m) and proper decay rates of the complexity terms. For the pretraining-related terms, the Rademacher complexity $\hat{\mathcal{R}}_{(n+m)/2}(\mathcal{F})$ scales as $O(\sqrt{R/(n+m)})$. Combined with the sample size condition $m+n = \text{poly}(d, 1/\Delta_{Y \rightarrow Y}, R)$, this ensures $\frac{cd}{\Delta_{Y \rightarrow Y}^2} \hat{\mathcal{R}}_{(n+m)/2}(\mathcal{F}) \leq \frac{cd}{\Delta_{Y \rightarrow Y}^2} \cdot C \sqrt{\frac{R}{n+m}} \rightarrow 0$. The term $\sqrt{(\log 2/\delta)/(n+m)}$ vanishes when δ is chosen as $\frac{1}{n}$. δ clearly converges to zero under this choice. For the downstream task terms, $n = \text{poly}(d, r_Y^I, 1/\Delta_{Y \rightarrow Y})$ ensures both components vanish: $r_Y^I C_d \sqrt{\frac{d}{n}} \rightarrow 0$ and $\sqrt{\frac{\log 1/\delta}{n}} = \sqrt{\frac{\log n}{n}} \rightarrow 0$. The polynomial growth conditions $m+n = \text{poly}(d, 1/\Delta_{Y \rightarrow Y}, R)$ and $n = \text{poly}(d, r_Y^I, 1/\Delta_{Y \rightarrow Y})$ guarantee all terms approach zero while maintaining $\Delta_{Y \rightarrow Y}, \Delta_d^{I \rightarrow Y} > 0$. This yields $\lim_{n, m \rightarrow \infty} \mathcal{E}^{I \rightarrow Y} = 0$. As $\mathcal{E}^{I \rightarrow Y} = \Pr_{G \sim \mathbb{P}_{\text{co}}, G_I \sim \mathcal{T}_I(\cdot|G)} [\mathbf{y}(G) \neq g_{f_Y^*, \hat{\mathbf{B}}}^{I \rightarrow Y}(G_I)]$, when labeled and unlabeled wild sets are large enough, we have $\mathcal{E}^{I \rightarrow Y} \approx \Pr_{G \sim \mathbb{P}_{\text{co}}, G_I \sim \mathcal{T}_I(\cdot|G)} [\mathbf{y}(G) \neq g_{f_Y^*, \hat{\mathbf{B}}}^{I \rightarrow Y}(G_I)] \approx 0$. $\square$

To justify invariant representations for reliable OOD detection, we recall two definitions and Theorem 4 from [14]:

Definition 7. For the population graph $G_{\text{pop}}^{I \rightarrow Y} = (\mathcal{G}_I, w^{I \rightarrow Y})$ and a subset $S_I \subseteq \mathcal{G}_I$, we define the Dirichlet conductance of S_I as

$$\phi_{G_{\text{pop}}^{I \rightarrow Y}}(S_I) = \frac{\sum_{G_I \in S_I, G'_I \notin S_I} w_{G_I G'_I}}{\sum_{G_I \in S_I} w_{G_I}}. \quad (24)$$

Definition 8. Let $G_{\text{pop}}^{I \rightarrow Y} = (\mathcal{G}_I, w^{I \rightarrow Y})$ be the graph of subgraphs. For an integer $i \in [2, |\mathcal{G}_I|]$, we define the sparsest i -partition as

$$\rho_i^{I \rightarrow Y} = \min_{S_{I,1}, \dots, S_{I,i}} \max \{\phi_{G_{\text{pop}}^{I \rightarrow Y}}(S_{I,1}), \dots, \phi_{G_{\text{pop}}^{I \rightarrow Y}}(S_{I,i})\}, \quad (25)$$

where $S_{I,1}, \dots, S_{I,i}$ are non-empty sets that form a partition of $\mathcal{G}_I$.

Theorem 4. Assume that the set of invariant subgraphs $\mathcal{G}_I$ is finite. Let $f_Y^* \in \arg \min_{f_Y: \mathcal{G}_I \rightarrow \mathbb{R}^d} \mathcal{L}^{I \rightarrow Y}$ be a minimizer of the spectral contrastive loss $\mathcal{L}^{I \rightarrow Y}$ in Eq. (8) with $d \in \mathbb{Z}^+$. Let $d' \geq r_Y$ be such that $d+1 = (1+\zeta)d'$, where $\zeta \in (0, 1)$. Then, there exists a linear probe $\mathbf{B}^{I \rightarrow Y*} \in \mathbb{R}^{d \times r_Y}$ and a universal constant c such that the error of the graph predictor $g_{f_Y^*, \mathbf{B}^{I \rightarrow Y*}}^{I \rightarrow Y}$ can be bounded by

$$\begin{aligned} & \Pr_{G \sim \mathbb{P}, G_I \sim \mathcal{T}_I(\cdot|G)} (g_{f_Y^*, \mathbf{B}^{I \rightarrow Y*}}^{I \rightarrow Y}(G_I) \neq \mathbf{y}(G)) \\ & \leq c \cdot \left(2\text{poly}\left(\frac{1}{\zeta}\right) \cdot \log(d+1) \cdot \frac{\phi_{\hat{\mathbf{B}}}^{I \rightarrow Y}}{\rho_{d'}^{I \rightarrow Y}} + \Delta_I^l(\mathbf{y}, \hat{\mathbf{y}}) + \Delta_I^u(\mathbf{y}, \hat{\mathbf{y}}) \right), \end{aligned} \quad (26)$$

where $g_{f_Y^*, \mathbf{B}^{I \rightarrow Y*}}^{I \rightarrow Y}(G) = \arg \max_{i \in [r_Y]} (f_Y^*(G)^\top \mathbf{B}^{I \rightarrow Y*})_i$ and $\rho_{d'}^{I \rightarrow Y}$ is the sparsest m -partition defined in Definition 8. Also, if we let $\lambda_i^{I \rightarrow Y}$ be the i -th smallest eigenvalue of the normalized Laplacian matrix of the population graph of invariant subgraphs, we can find a matrix $\mathbf{B}^{I \rightarrow Y*}$ satisfying the above equations with norm bound $\|\mathbf{B}^*\|_F \leq \frac{1}{1-\lambda_d^{I \rightarrow Y}}$.

The following lemma connects $\phi_{\hat{\mathbf{B}}}^{I \rightarrow Y}$ with Assumption 1, enabling simplification of Theorem 4.

Lemma 2. Let $G_{\text{pop}}^{I \rightarrow Y} = (\mathcal{G}_I, w^{I \rightarrow Y})$ be the population graph of invariant subgraphs, r_Y be the number of underlying classes. Let $S_1^{I \rightarrow Y}, S_2^{I \rightarrow Y}, \dots, S_{r_Y}^{I \rightarrow Y}$ be the partition induced by the classifier g_Y

in Assumption 1. Then, there exists an extended labeling function $\hat{\mathbf{y}}$ such that $\phi_{\hat{\mathbf{B}}}^{I \rightarrow Y} \leq 2 \left(\sum_{i \in \mathcal{Y}} \Delta_I^l(\mathbf{y}, \hat{\mathbf{y}}) + \Delta_I^u(\mathbf{y}, \hat{\mathbf{y}}) \right) = 0$.

PROOF. We begin by expanding the expression for $\phi_{\hat{\mathbf{B}}}^{I \rightarrow Y}$:

$$\begin{aligned} \phi_{\hat{\mathbf{B}}}^{I \rightarrow Y} &= \sum_{G_I, G'_I \in \mathcal{X}} w_{G_I G'_I} \cdot \mathbb{1} [\hat{\mathbf{y}}(G_I) \neq \hat{\mathbf{y}}(G'_I)] \\ &= \sum_{G_I, G'_I \in \mathcal{Y}} \mathbb{E}_{G_I \sim \mathbb{P}_{I_i}} \mathbb{E}_{G'_I \sim \mathbb{P}_{I_i}} [\mathcal{T}_I(G_I|G_I) \mathcal{T}_I(G'_I|G'_I) \cdot \mathbb{1} [\hat{\mathbf{y}}(G_I) \neq \hat{\mathbf{y}}(G'_I)]] \\ &\quad + \sum_{G_I, G'_I} \mathbb{E}_{G_u \sim \mathbb{P}_u} [\mathcal{T}_I(G_I|G_u) \mathcal{T}_I(G'_I|G_u) \cdot \mathbb{1} [\hat{\mathbf{y}}(G_I) \neq \hat{\mathbf{y}}(G'_I)]] . \end{aligned} \quad (27)$$

Next, we utilize the following inequality:

$$\mathbb{1} [\hat{\mathbf{y}}(G_I) \neq \hat{\mathbf{y}}(G'_I)] \leq \mathbb{1} [\hat{\mathbf{y}}(G_I) \neq \mathbf{y}(G)] + \mathbb{1} [\hat{\mathbf{y}}(G'_I) \neq \mathbf{y}(G)] . \quad (28)$$

Applying this inequality to the previous expansion of $\phi_{\hat{\mathbf{B}}}^{I \rightarrow Y}$, we obtain:

$$\begin{aligned} \phi_{\hat{\mathbf{B}}}^{I \rightarrow Y} &\leq 2 \sum_{i \in \mathcal{Y}} \mathbb{E}_{G_I \sim \mathbb{P}_{I_i}} \left[\sum_{G_I} \mathcal{T}_I(G_I|G_I) \cdot \mathbb{1} [\hat{\mathbf{y}}(G_I) \neq \mathbf{y}(G_I)] \right] \\ &\quad + 2 \mathbb{E}_{G_u \sim \mathbb{P}_u} \left[\sum_{G_I} \mathcal{T}_I(G_I|G_u) \cdot \mathbb{1} [\hat{\mathbf{y}}(G_I) \neq \mathbf{y}(G_u)] \right] \\ &= 2 \left(\sum_{i \in \mathcal{Y}} \Delta_I^l(\mathbf{y}, \hat{\mathbf{y}}) + \Delta_I^u(\mathbf{y}, \hat{\mathbf{y}}) \right) = 0. \end{aligned} \quad (29)$$

$\square$

In Lemma 3, we theoretically analyze OOD detection via linear probing error—a standard tool in representation learning theory [5, 31]. In practice, with semantic-shifted OOD graphs unobserved, we adopt a KNN distance metric as the detection criterion.

Lemma 3. In the setting of Theorem 4, given invariant subgraphs $G_I \sim \mathcal{T}_I(\cdot|G)$, let $f_Y^*(\cdot)$ be the optimal GNN minimizing Eq. (8). Then there exists a linear probe $\mathbf{B}^{I \rightarrow Y*}$ yielding a perfect OOD detector $h_{f_Y^*, \mathbf{B}^{I \rightarrow Y*}}^{I \rightarrow Y}$:

$$\Pr_{G \sim \mathbb{P}_{I \cup \mathbb{P}_{\text{co}}}} [h_{f_Y^*, \mathbf{B}^{I \rightarrow Y*}}^{I \rightarrow Y}(G) \neq \text{IN}] + \Pr_{G \sim \mathbb{P}_{\text{se}}} [h_{f_Y^*, \mathbf{B}^{I \rightarrow Y*}}^{I \rightarrow Y}(G) \neq \text{OUT}] = 0. \quad (30)$$

PROOF. By applying Theorem 4, we obtain the following error bound:

$$\begin{aligned} & \Pr_{G \sim \mathbb{P}, G_I \sim \mathcal{T}_I(\cdot|G)} (g_{f_Y^*, \mathbf{B}^{I \rightarrow Y*}}^{I \rightarrow Y}(G_I) \neq \mathbf{y}(G)) \\ & \leq c \cdot \left(2\text{poly}\left(\frac{1}{\zeta}\right) \cdot \log(d+1) \cdot \frac{\phi_{\hat{\mathbf{B}}}^{I \rightarrow Y}}{\rho_{d'}^{I \rightarrow Y}} + \Delta_I^l(\mathbf{y}, \hat{\mathbf{y}}) + \Delta_I^u(\mathbf{y}, \hat{\mathbf{y}}) \right), \end{aligned} \quad (31)$$

where $\mathbf{B}^{I \rightarrow Y*} \in \mathbb{R}^{d \times r_Y}$ is the original classifier for all r_Y classes (including both IN and OUT categories). By Lemma 2, $\Delta_I^l(\mathbf{y}, \hat{\mathbf{y}}) = 0$ and $\Delta_I^u(\mathbf{y}, \hat{\mathbf{y}}) = 0$, we derive $\Pr_{G \sim \mathbb{P}, G_I \sim \mathcal{T}_I(\cdot|G)} (g_{f_Y^*, \mathbf{B}^{I \rightarrow Y*}}^{I \rightarrow Y}(G_I) \neq \mathbf{y}(G)) \leq 0$. We then construct the detection probe through linear transformation. Define $\mathbf{B}^* = \mathbf{B}^{I \rightarrow Y*} \mathbf{T}$ where $\mathbf{T} \in \mathbb{R}^{k \times 2}$ is a projection matrix that maps IN-class logits to the IN dimension and maps OUT-class logits to the OUT dimension. This construction preserves the separation between IN and OUT distributions, yielding:

$$\begin{aligned} & \Pr_{G \sim \mathbb{P}_{I \cup \mathbb{P}_{\text{co}}, G_I \sim \mathcal{T}_I(\cdot|G)}} (h_{f_Y^*, \mathbf{B}^*}^{I \rightarrow Y}(G_I) \neq \text{IN}) \\ & + \Pr_{G \sim \mathbb{P}_{\text{se}}, G_I \sim \mathcal{T}_I(\cdot|G)} (h_{f_Y^*, \mathbf{B}^*}^{I \rightarrow Y}(G_I) \neq \text{OUT}) \leq 0, \end{aligned} \quad (32)$$

where $h_{f_Y^*, \mathbf{B}^*}^{I \rightarrow Y}(G) = \arg \max_{i \in \{\text{IN}, \text{OUT}\}} (f_Y^*(G)^\top \mathbf{B}^*)_i$. $\square$

Table 5: Averaged training and inference time (sec.) per epoch of various methods on GD-Motif.

Methods	Energy	Scone	Bridge	UniGOOD
Training	5.49	5.77	7.46	10.88
Inference	0.10	0.10	0.09	0.22

Theorem 5 establishes that spectral contrastive learning on invariant subgraphs simultaneously guarantees both OOD generalization and detection capabilities, i.e., the solution to Problem 1. A direct proof combines Lemma 1 with Lemma 3.

Theorem 5. *In the setting of Lemma 1 and 3, given invariant subgraphs $G_I \sim \mathcal{T}_I(\cdot|G)$. For any $f_Y^*(\cdot)$ be the optimal GNN minimizing Eq. (8) and a linear head $\hat{\mathbf{B}}^{I \rightarrow Y}$ in Lemma 1, $g_{f_Y^*, \mathbf{B}^{I \rightarrow Y^*}}$ achieves satisfactory performance on covariate-shifted OOD graphs, i.e., minimizes $\Pr_{G \sim \mathbb{P}_{\text{co}}} [y(G) \neq g_{f_Y^*, \mathbf{B}^{I \rightarrow Y^*}}(G)]$, and there exists a linear probe $\mathbf{B}^{I \rightarrow Y}$ yielding a perfect OOD detector $h_{f_Y^*, \mathbf{B}^{I \rightarrow Y^*}}$, which minimizes $\Pr_{G \sim \mathbb{P}_I \cup \mathbb{P}_{\text{co}}} [h_{f_Y^*, \mathbf{B}^{I \rightarrow Y^*}}(G) \neq \text{IN}] + \Pr_{G \sim \mathbb{P}_{\text{co}}} [h_{f_Y^*, \mathbf{B}^{I \rightarrow Y^*}}(G) \neq \text{OUT}]$.*

A.3 Invariant Subgraph Discovery Guarantee

First, we formalize the assumption that positive pair and negative pair items in contrastive learning loss are affected by the invariant subgraph distribution $\hat{\mathcal{T}}_I(\cdot|G)$:

Assumption 2. *Let $\Phi(\cdot) = \hat{\mathcal{T}}_I(\cdot|G)$ denote the subgraph generator that models the invariant subgraph distribution from an input graph G . We assume that the graph encoder $f_Y(\cdot)$ satisfies the following bounds: $\mathcal{L}_1^{I \rightarrow Y} \leq 1 - \epsilon_p(\hat{\mathcal{T}}_I)$, $\mathcal{L}_2^{I \rightarrow Y} \leq 1 - \epsilon_p(\hat{\mathcal{T}}_I)$, $\mathcal{L}_3^{I \rightarrow Y} \geq \epsilon_n(\hat{\mathcal{T}}_I)$, $\mathcal{L}_4^{I \rightarrow Y} \geq \epsilon_n(\hat{\mathcal{T}}_I)$, and $\mathcal{L}_5^{I \rightarrow Y} \geq \epsilon_n(\hat{\mathcal{T}}_I)$, where $\epsilon_p : \mathcal{T} \rightarrow [0, 1]$ is a strictly concave function quantifying the inevitable information loss during subgraph extraction and $\epsilon_n : \mathcal{T} \rightarrow [0, 1]$ is a strictly concave function representing the fundamental similarity floor due to shared substructures.*

Through Assumption 2, we have $\mathcal{L}^{I \rightarrow Y} \geq 4\epsilon_p(\hat{\mathcal{T}}_I) + 4\epsilon_n(\hat{\mathcal{T}}_I) - 4$. Similarly, $\mathcal{L}^{V \rightarrow Y} \geq 2\epsilon_p(\hat{\mathcal{T}}_V) - 2$, $\mathcal{L}^{I \rightarrow E} \geq 2\epsilon_p(\hat{\mathcal{T}}_I) - 2$, and $\mathcal{L}^{V \rightarrow E} \geq 2\epsilon_p(\hat{\mathcal{T}}_V) + 2\epsilon_n^{V \rightarrow E} - 2$.

Theorem 1. *There exists an optimal invariant subgraph generator $\Phi^*(G) = \mathcal{T}_I(\cdot|G)$ and its complement $\Psi^*(G) = G \setminus \Phi^*(G) = \mathcal{T}_V(\cdot|G)$ if and only if it simultaneously minimizes $\mathcal{L}^{I \rightarrow Y}$, $\mathcal{L}^{V \rightarrow Y}$, $\mathcal{L}^{I \rightarrow E}$, and $\mathcal{L}^{V \rightarrow E}$.*

PROOF. ($\Rightarrow$) If there exists an optimal invariant subgraph generator $\Phi^*(G) = \mathcal{T}_I(\cdot|G)$, we can construct a graph encoder $f_Y^*(\cdot)$ with the following ideal properties: (1) For any invariant subgraphs G_I, G_I^+ derived from graphs of the same class or the same unlabeled input graph, their embeddings satisfy: $f_Y^*(G_I) = f_Y^*(G_I^+)$. This guarantees maximum similarity for positive pairs: $\mathcal{L}_1^{I \rightarrow Y} = \mathcal{L}_2^{I \rightarrow Y} = 1$. (2) For any invariant subgraphs G_I, G_I^- derived from different classes or different unlabeled graphs, their embeddings satisfy: $\langle f_Y^*(G_I), f_Y^*(G_I^-) \rangle = 0$. Thus achieving minimum similarity for negative pairs: $\mathcal{L}_3^{I \rightarrow Y} = \mathcal{L}_4^{I \rightarrow Y} = \mathcal{L}_5^{I \rightarrow Y} = 0$. The contrastive loss then attains its theoretical minimum: $\mathcal{L}^{I \rightarrow Y} = -4$. Similarly,

we can demonstrate optimality for other loss terms: $\mathcal{L}^{V \rightarrow Y} = -2$, $\mathcal{L}^{I \rightarrow E} = -2$, and $\mathcal{L}^{V \rightarrow E} = -2$.

($\Leftarrow$) If the four contrastive learning losses reach their minimum values, i.e., $\mathcal{L}^{I \rightarrow Y} = -4$, $\mathcal{L}^{V \rightarrow Y} = -2$, $\mathcal{L}^{I \rightarrow E} = -2$, and $\mathcal{L}^{V \rightarrow E} = -2$, we have $\epsilon_p(\hat{\mathcal{T}}_I) = \epsilon_n(\hat{\mathcal{T}}_I) = 0$. Given that ϵ_p and ϵ_n is a strictly concave function, and there exists an optimal invariant subgraph generator $\Phi^*(G) = \mathcal{T}_I(\cdot|G)$ satisfying $\epsilon_p(\mathcal{T}_I) = \epsilon_n(\mathcal{T}_I) = 0$. We conclude that the current invariant subgraph generator $\Phi(\cdot)$ must be optimal, i.e., $\Phi(G) = \hat{\mathcal{T}}_I(\cdot|G) = \mathcal{T}_I(\cdot|G) = \Phi^*(G)$. $\square$

Having established in Theorem 1 that our four objectives enable accurate invariant subgraph identification, we now analyze how they collectively guarantee the invariance principle. Let $e(\cdot)$ be the environment labeling function. For the population graphs $G_{\text{pop}}^{V \rightarrow Y}$, we define the independent class separate from $\mathcal{Y}$ as y_V . For the population graphs $G_{\text{pop}}^{I \rightarrow E}$, we define the independent class as e_I .

Theorem 6. *Let the subgraphs $\hat{G}_I, \hat{G}_I' \sim \hat{\mathcal{T}}_I(\cdot|G)$ and the complements $\hat{G}_V, \hat{G}_V' \sim \hat{\mathcal{T}}_V(\cdot|G)$ such that there exists a graph classifier $g_Y^*(\cdot)$ that can predict $y(G)$ given $\hat{G}_I$ ($G_{\text{pop}}^{I \rightarrow Y}$) and predict y_V given $\hat{G}_V$ ($G_{\text{pop}}^{V \rightarrow Y}$) with error at most 0 and there exists an environment classifier $g_E^*(\cdot)$ that can predict $e(G)$ given $\hat{G}_V$ ($G_{\text{pop}}^{V \rightarrow E}$) and predict e_I given $\hat{G}_I$ ($G_{\text{pop}}^{I \rightarrow E}$) with error at most 0. Then, $\hat{G}_I$ are the true invariant subgraphs G_I .*

PROOF. From the assumption about $G_{\text{pop}}^{I \rightarrow Y}$ and $G_{\text{pop}}^{V \rightarrow Y}$ that $g_Y^*(\hat{G}_I) = y(G)$ and $g_Y^*(\hat{G}_V) = g_Y^*(\hat{G}_V') = y_V$, we derive $P(Y|G_V) = P(g_Y^*(G_I)|G_V) = P(g_Y^*(G_I)) = P(Y)$, so $G_V \perp Y$. Similarly, for the other two population graphs, we have $G_I \perp E$. Next, following [12], we use $\hat{G}_V \perp Y$ and $\hat{G}_I \perp E$ to prove $G_V \subseteq \hat{G}_V$ and $\hat{G}_I \subseteq G_I$.

Assume G_V is not a subgraph of $\hat{G}_V$. Then there exists non-trivial $G_q \subseteq G_V$ such that $G_q \not\subseteq \hat{G}_V$. Since invariant subgraphs determine Y , G_q is not independent of Y . This contradicts $\hat{G}_V \perp Y$, proving $G_V \subseteq \hat{G}_V$. Similarly, $\hat{G}_I \subseteq G_I$ follows from $\hat{G}_I \perp E$. Given that $\hat{G}_V = G \setminus \hat{G}_I$ and $G_V = G \setminus G_I$, we conclude that $\hat{G}_I = G_I$. $\square$

Theorem 6 shows that vanishing generalization errors on all four population graphs guarantee $G_V \perp Y$ and $G_I \perp E$, thus ensuring precise invariant subgraph identification.

B Time Complexity Analysis

UniGOOD has a time complexity of $O((|\mathcal{E}| + |\mathcal{B}|^2)d + |\mathcal{V}|d^2)$, where $|\mathcal{V}|$, $|\mathcal{E}|$, $|\mathcal{B}|$, and d denote the numbers of nodes, edges, batch size, and representation dimensionality, respectively. It avoids spectral decomposition in practice, using it only as a theoretical tool. The cost breaks down into: (1) $O(|\mathcal{E}|d + |\mathcal{V}|d^2)$ for GNN operations and subgraph modeling, and (2) $O(|\mathcal{B}|^2d)$ for the optimization objective. Compared to standard GNNs ($O(|\mathcal{E}|d + |\mathcal{V}|d^2)$), UniGOOD is highly efficient when $|\mathcal{B}|^2 \ll |\mathcal{E}|$.

C Running Time Analysis

We compare the average training/inference times of UniGOOD and baselines on GD-Motif. As shown in Table 5, UniGOOD maintains computational efficiency comparable to other methods, with only marginal overhead from its invariance regularizer. This result is consistent with our time complexity analysis in Section B.